UM-2 analysis

The x_0 mass mechanism, and the validity of associating dark matter and dark energy with the dirac16complex field

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1 How to read this document

The author asked for two analyses: of the mechanism that gives every non-massless particle its mass through the dependence of its wave function on the hidden space coordinate x_0 (periodic with period $\pi/(6H)$, the uncertainty relation $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$ being the physical reason), and of the validity of associating dark matter and dark energy with dirac16complex. This document carries out both within UM-2. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD, ASSUMED, INVENTED, DERIVED (computed and verified), ESTIMATE, FROM MEMORY (a measured bound quoted from memory), OPEN. Units $\hbar = c = 1$. Coordinates are the author's $x_0, \ldots, x_7$ (x_0 hidden space, x_1, x_2, x_3 space, x_4 time, x_5, x_6, x_7 extra times); W^+ has charge $+e$, W^- charge $-e$.

Symbols with more than one use: $\xi = y/L$ is the scaled hidden coordinate (primes denote $d/d\xi$), t the cosmic time in the bounds on varying constants; $\Lambda = (m\,f\,L)^2$ is a dimensionless eigenvalue of an x_0 operator, while in the

formula for the four-dimensional cosmological term Λ_4 the symbol $\Lambda = -10H^2$ is the eight-dimensional cosmological constant; $\Lambda_\star$ is the scale of the effective interactions; k is the momentum of a mode along x_1, x_2, x_3 in h_j and in $\varepsilon = e^{-a_4}|k|$, and the block variable $1/f^2$ in k_*, $G(k)$ and Λ_4; ρ is an energy density (ρ_0 the electroweak parameter) and ϱ a probability density in ξ; w is an equation-of-state parameter (w_0, w_a of the form $w = w_0 + w_a(1-a)$, a the scale factor), except in the fermion mass condition ($w = \sqrt{c^2 - e^2}$) and the Lovelock weight w_2; $z = 6Hx_0$, except where it is a redshift.

Summary of the verdicts.

question	verdict
Can every mass come from the x_0 dependence of wave functions periodic with period $\pi/(6H)$?	YES (DERIVED): every four-dimensional mass of UM-2 is an eigenvalue of an x_0 operator; graviton, photon, gluons and the dirac16complex fold band are exactly massless
Is the uncertainty relation the reason for the masses?	ONLY FOR THE HEAVY STATES: it bounds the Kaluza-Klein and odd excitations (and the dark matter) from below, but the light masses ($W^\pm$, Z, e, ν, u, d) lie 10^{-3} to 10^{-16} below the uncertainty estimate and come from the tip of the x_0 cell
Do the extra times deflate to irrelevance?	IN THE PRIMORDIAL ERA, then FROZEN: they deflate by up to a factor 2.9 into the stabilised block; a continuing deflation is excluded because every mass and coupling depends on the block size
Is dirac16complex dark matter?	CONDITIONALLY YES: the lightest odd quantum of the Ψ_B half (44.3 TeV, cold, neutral, stable) is a viable candidate whose relic abundance fixes one portal scale; the massless fold bands are radiation, not dark matter
Is dirac16complex dark energy?	NOT SUPPORTED: no positive-energy state has $w < 0$; a negative-coupling condensate runs away; the dark energy of UM-2 is a tuned four-dimensional cosmological term

2 The x_0 mass mechanism

2.1 The author's requirement in equations

The hidden coordinate enters through $z = 6Hx_0$; one period is $\ell_0 = \pi/(6H)$; on each half of a period

$$y = \frac{\ln|\sin z|}{6H} \in (-L, 0], \qquad dy = \cot z \, dx_0, \qquad ds_{hidden} = f \, dy,$$

the fold $y = 0$ ($z = \pi/2$) joining the two halves and the tip $y = -L$ ($\eta = HL = 3$) closing them. Every wave function of UM-2 depends on x_0 through $y(z)$, so it is periodic in x_0 with period $\pi/(6H)$. The uncertainty relation, in the proper hidden length $s = fy$, reads $\Delta s \, \Delta p_s \geq \frac{1}{2}$ on an unbounded line (on the cell, which ends at the tips, it acquires a tip term; section 2.3), and the author's estimate of a mass is

$$m_H = \frac{\hbar c}{2\Delta s}, \qquad \Delta s = fL \, \Delta\xi, \qquad \Delta\xi^2 = \frac{\int_{-1}^{0} \xi^2 \varrho(\xi) \, d\xi}{\int_{-1}^{0} \varrho(\xi) \, d\xi},$$

the root-mean-square proper distance from the fold of the probability density ϱ in the flat measure ($\varrho = \tilde{u}^2$, $\tilde{u} = e^{3\eta\xi/2}u$, for bosons; $\varrho = |\chi|^2$ for fermions), $\xi = y/L$; over the full period the density is symmetric about the fold, so $\Delta\xi$ is also the spread of the full-period profile.

2.2 Every mass is an x_0 eigenvalue (DERIVED)

With $\xi = y/L$:

- **Gauge bosons and the graviton**: $-e^{-3\eta\xi}(e^{3\eta\xi}u')' = \Lambda u$, $\Lambda = (m\,fL)^2$; fold $u'(0) = 0$ (even) or $u(0) = 0$ (odd); tip $u'(-1) = \beta u(-1)$ with $\beta = 0$ for the photon, the gluons and the graviton and $\beta_W = 0.011932$ (W^+, W^-), $\beta_Z = \beta_W/\cos^2\theta_W$ (Z; on-shell angle, $\cos^2\theta_W = m_W^2/m_Z^2 = 0.77666$).
- **Fermions**: $h_j = j[-\frac{i}{f}\sigma_1\partial_y + \mu\sigma_2 + e^{-a_4 k}\sigma_3]$ (k the momentum) with the chiral boundary conditions and the tip mixing $b_L = \kappa_f b_R$, $a_R = -\kappa_f a_L$; the mass is the root of $D(e) = e^2 s_L s_R - \kappa_f^2(C_L - c_L s_L)(C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$, $c = f\mu L$.
- **dirac16complex**: h_j with $M = m$, fold $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$.

The masses (DERIVED): photon, gluons, graviton 0 exactly (flat profile); dirac16complex fold band 0 exactly (chiral zero mode, $\varepsilon = e^{-a_4|k|}$); $W^\pm$ 80.36 GeV and Z 91.17 GeV (fit of β_W to ρ_0); ν, e, u, d with $c = 18.67, 10.30, 9.54, 9.13$; the dark matter $3.102204\,m = 44.31$ TeV; Kaluza-Klein levels from 121 TeV; every odd mode and every excitation of the gauge fields and the graviton above the gap $3H/(2f_*) = 99.4$ TeV; the Ψ excitations from its lowest odd level, 44.3 TeV, upwards.

2.3 What the uncertainty relation does and does not bound (DERIVED)

For a boson, write $u = e^{-3\eta\xi/2}\tilde{u}$. The exact identity

$$e^{3\eta\xi}u'^2 = \tilde{u}'^2 + \frac{9}{4}\eta^2\tilde{u}^2 - \frac{3}{2}\eta\,(\tilde{u}^2)'$$

turns the Rayleigh quotient $(m\,fL)^2 = \big[\int e^{3\eta\xi}u'^2 d\xi + \beta e^{-3\eta}u(-1)^2\big]/\int e^{3\eta\xi}u^2 d\xi$ into the decomposition

$$(m\,fL)^2 = \langle p^2\rangle + \frac{9\eta^2}{4} - \frac{3\eta}{2}\varrho(\text{fold}) + \Big(\frac{3\eta}{2} + \beta\Big)\varrho(\text{tip}), \qquad \langle p^2\rangle = \frac{\int \tilde{u}'^2}{\int \tilde{u}^2}, \quad \varrho = \frac{\tilde{u}^2}{\int \tilde{u}^2},$$

and for a fermion $(\varepsilon fL)^2 = \langle|\chi'|^2\rangle + c^2 - c\,[\chi^\dagger\sigma_3\chi]_{-1}^0/\int|\chi|^2$, with $\langle p^2\rangle = \langle|\chi'|^2\rangle$ and $\varrho = |\chi|^2/\int|\chi|^2$ (integrals over $\xi \in [-1, 0]$). Only the hidden kinetic term is bounded by the uncertainty relation. On the cell, which ends at the tips, the relation reads

$$\sqrt{\langle p^2\rangle}\,\Delta\xi \geq \frac{1 - \varrho(\text{tip})}{2},$$

because $1 = \int_{-1}^0 \varrho\,d\xi = \varrho(\text{tip}) - \int_{-1}^0 \xi\varrho'\,d\xi$ (integration by parts; the fold end $\xi = 0$ contributes nothing) and the Cauchy-Schwarz inequality gives $|\int \xi\varrho'\,d\xi| \leq 2\Delta\xi\sqrt{\langle p^2\rangle}$. The familiar form $\langle p^2\rangle \geq 1/(4\Delta\xi^2)$ needs a profile negligible at the tip: the broad fold zero mode of Ψ violates it ($\langle p^2\rangle\Delta\xi^2 = 0.097$, $\varrho(\text{tip}) = 0.489$) and satisfies the relation with the tip term. Computed for every class of particle (DERIVED; sum rules reproduce the eigenvalues to 10^{-8}, the dark matter to 10^{-11}; the relation with the tip term holds for every state, with the smallest margin 0.057 for the Ψ fold band):

Bosons (the warp term is $9\eta^2/4 = 20.25$ for every mode):

state	$\langle p^2\rangle$	fold term	tip term	$(m\,fL)^2$	m/m_H
photon, gluons, graviton	20.250	-40.505	0.0050	0	0
$W^\pm$	20.250	-40.505	0.0050	1.32×10^{-5}	1.14×10^{-3}
Z	20.250	-40.505	0.0050	1.70×10^{-5}	1.29×10^{-3}
first even Kaluza-Klein mode	9.870	-2.949	2.949	30.120	7.49
first odd mode	4.891	0	1.948	27.089	6.47

Fermions (the kinetic term $\langle|\chi'|^2\rangle$ equals the bulk term c^2 for the zero modes):

state	$\langle\lvert\chi'\rvert^2\rangle$	bulk c^2	boundary term	$(\varepsilon f L)^2$	m/m_H
ν zero mode	348.60	348.60	-697.21	0	1.7×10^{-16}
e zero mode	106.17	106.17	-212.34	0	3.2×10^{-9}
u zero mode	91.10	91.10	-182.21	0	1.4×10^{-8}
d zero mode	83.43	83.43	-166.86	0	3.3×10^{-8}
Ψ fold band	0.418	0.418	-0.836	0	0
Ψ_B dark matter	2.608	0.418	0.998	4.024	2.29

(the masses of ν, e, u, d come from the tip mixing, not from these zero modes). Proper spreads, uncertainty estimates, tip densities and uncertainty products:

state	Δs (m)	$m_H = \hbar c/(2\Delta s)$ (GeV)	ϱ(tip)	$\langle p^2\rangle\Delta\xi^2$
photon, gluons, graviton; $W^\pm$; Z	1.40×10^{-21}	70 500	1.1×10^{-3}	0.497
first even Kaluza-Klein mode; first odd mode	6.09; 5.56×10^{-21}	16 191; 17 756	0.655; 0.433	4.59; 1.89
ν; e; u; d zero modes	3.38; 6.13; 6.62; 6.92×10^{-22}	291 611; 160 929; 149 075; 142 660	2.3×10^{-15} to 2.1×10^{-7}	0.500
Ψ fold band; Ψ_B dark matter	4.31; 5.09×10^{-21}	22 884; 19 378	0.489; 0.772	0.097; 0.847

Three classes follow.

- **Massless although localised.** The photon, the gluons and the graviton are localised at the fold ($\Delta s = 1.40 \times 10^{-21}$ m, $m_H = 70.5$ TeV, $\langle p^2\rangle\Delta\xi^2 = 0.497$), yet exactly massless: the fold term cancels the hidden kinetic energy and the warp. The cancellation is exact because the flat profile is an eigenfunction with $\Lambda = 0$, protected by four-dimensional gauge invariance and general covariance. The fold bands of dirac16complex are massless for the analogous chiral reason ($\chi = (e^{f\int M dy}, 0)$ is a zero mode for every mass profile).
- **Heavy: the uncertainty relation is a true lower bound.** Every x_0 excitation of the gauge fields and the graviton lies above the gap of their operator: for every tip coupling β from 0 to 10^6 exactly one even level lies below $9\eta^2/4$ (the zero mode, lifted only by the tip) and no odd level does (the gap theorem, DERIVED), so every odd mode and every excitation of these fields has $m \geq 3H/(2f_*) = 99.4$ TeV. The Dirac operator of Ψ has no such warp gap; its excitations start at the lowest odd level, $3.10m = 44.3$ TeV (the dark matter). For all these states m_H is exceeded by factors 2.3 to 7.5 (dark matter 2.29, first odd 6.47, first even 7.49): the author's Heisenberg estimate holds as a bound, and the dark-matter mass is an x_0 mass in this sense (its hidden wave number p carries $p^2/\varepsilon^2 = 0.896$ of ε^2).
- **Light: the mass comes from the tip, not from localisation.** For $W^\pm$, Z, e, ν, u, d the x_0 dependence that costs energy sits at the tip, where the probability is small: $m/m_H = 10^{-3}$ ($W^\pm$, Z) to 10^{-16} (ν). For the fermions a sharper localisation gives a LIGHTER particle, $d\ln m_f/d\ln c = -36.3, -19.6, -18.1, -17.3$ (ν, e, u, d): the reverse of the Heisenberg intuition.

2.4 Proper versus coordinate localisation (DERIVED)

The photon zero mode, localised at the fold in proper length, has 61.9 % of its probability within $\pi/8$ of the tips in the coordinate $z = 6Hx_0$: localisation in the coordinate x_0 is chart dependent, because $ds_{hidden} = f\cot z\, dx_0$ vanishes at the fold and diverges at the tips. A profile confined to a small proper distance from the fold ($\Delta\xi = 0.157$, i.e. $|y| \lesssim 0.47/H$, $\sin z \gtrsim e^{-2.8}$) therefore extends over almost the whole coordinate range of z, and its probability per unit z is largest near the tips. Only the proper hidden length enters the masses; a statement "Δx_0 small" must therefore be read as "Δs small".

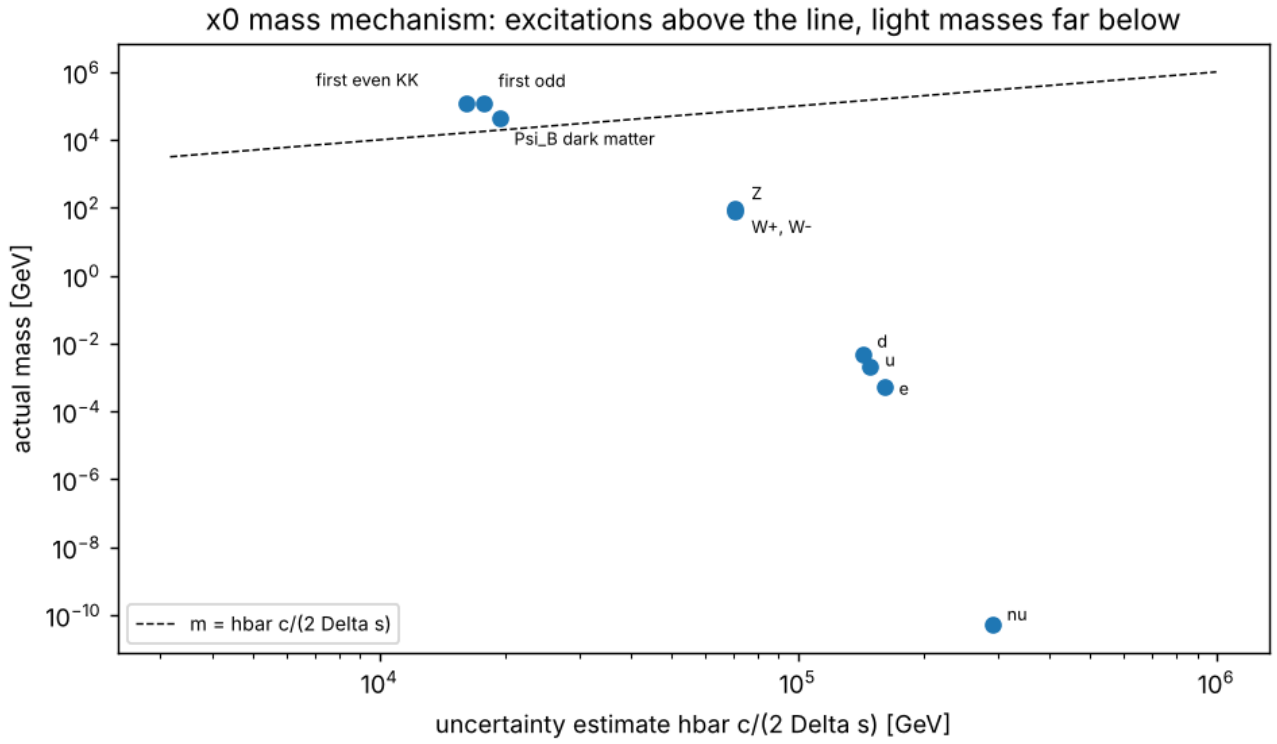


Figure 1: The x_0 mass mechanism: actual masses against the uncertainty estimate $m_H = \hbar c/(2\Delta s)$; the excitations and the dark matter lie above the line, the light particles far below.

2.5 The mass mechanism, the block and the deflating extra times

Every mass and coupling depends on the block scale f (DERIVED):

quantity X	$d\ln X/d\ln f$
Kaluza-Klein masses, the gap $3H/(2f)$	-1
m_W (and m_Z)	-0.501
m_e, m_u, m_d, m_ν	-20.6, -19.1, -18.3, -37.3
M_{DM}	-0.772
α and the gauge couplings ($g^2 = \hat{g}^2/V_\perp \propto f^{-4}$)	-4
G_N ($8\pi G_N = \kappa/(G(k)V_\perp)$: $-4 + 2k_*G'(k_*)/G(k_*)$)	$+2.05$

The author's metric lets the extra times x_5, x_6, x_7 (and, in the corrected metric, x_0 with them) deflate exponentially. In UM-2 they do so in the primordial era: the full nonlinear equations take a large block into the stabilised one at rates up to $|h| = 2.63H$ (in the stored runs), deflating by up to $f(k_c)/f_* = 2.917$, after which the block oscillates about f_* (the radion) and stays frozen. A continuing deflation today is excluded (ESTIMATE, bounds FROM MEMORY): the atomic-clock bound $|d\ln\alpha/dt| < 10^{-17}$/yr ($t$ the cosmic time) and $d\ln\alpha/d\ln f = -4$ require $|h| < 2.5\times10^{-18}$/yr $= 3.6\times10^{-8}H_0$; lunar ranging ($|d\ln G/dt| < 10^{-13}$/yr, $d\ln G_N/d\ln f = +2.05$) requires $|h| < 7.1\times10^{-4}H_0$; a block deflating at the Hubble rate would change the electron mass by 1.4×10^{-9} per year. The extra times are irrelevant at low energies not because they keep deflating but because they are small (the proper half cell $f_*L = 3f_*/H = 8.9\times10^{-21}$ m, Kaluza-Klein masses above 100 TeV) and frozen. The price is the stabiliser (ASSUMED, origin OPEN) and the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi\bar{M}_{Pl}^2)$, with the reduced Planck mass $\bar{M}_{Pl} = 2.435\times10^{18}$ GeV and an order-one coupling factor C_r, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma\bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem, OPEN).

2.6 Assessment of the mechanism

The author's mechanism works in UM-2 in a precise sense: all masses are x_0 eigenvalues, the period $\pi/(6H)$ is built into every wave function, and the massless particles are exactly those protected by symmetry. The uncertainty relation, however, is the reason for the masses of the heavy states only. The light masses need two further ingredients that are not consequences of the uncertainty relation: the tip of the cell (where electroweak symmetry is broken by Σ) and exponentially localised chiral zero modes, whose overlap at the tip ($\zeta(c)^2 \simeq 2c\,e^{-2c}$) produces the hierarchy from order-one bulk masses. The scales are not free: once the Kaluza-Klein scale is fitted ($m_{KK} = 121$ TeV from ρ_0), the dark-matter mass (44 TeV) and the radion (48 TeV) follow.

3 dirac16complex as dark matter

3.1 The candidate

Ψ splits into two four-dimensionally Lorentz-invariant halves, Ψ_A ($B=-1$) and Ψ_B ($B=+1$), $B = -iC\gamma^{x_4}$. In the good sector every term of UM-2 commutes with B (DERIVED with the author's matrices: B commutes with C, with γ^a and $C\gamma^a$ for $a = x_0, \ldots, x_4$, anticommutes with them for $a = x_5, x_6, x_7$, and $\gamma^M\Omega_M = \frac{3}{2}\frac{H}{f}\gamma^{x_0} + (\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2})\gamma^{x_4}$ contains γ^{x_0} and γ^{x_4} only), so N_A and N_B are separately conserved. The tip projector of the Z portal, $P_+ = (1+\Gamma_4)/2 = (1 - B\gamma^{x_0})/2$ ($\Gamma_4 = -B\gamma^{x_0}$, DERIVED), annihilates every Ψ_B orbital at the tip. The dark-matter candidate is the lightest odd state of Ψ_B:

- a Dirac fermion of mass $M_{DM} = \sqrt{m^2+p^2} = 3.102204\,m = 44.31$ TeV ($\tan(p\,fL) = -p/m$, $m = H$);
- cold: $w = p_3/\rho = 0$ for quanta at rest (Kohn-Sham, DERIVED; 2.8×10^{-5} with the benchmark self-coupling);
- electrically neutral (a gauge singlet) and with no Z coupling and no transition current at the tip (exactly);
- stable: N_B is conserved and its fold parity is exact (no tip operator reaches Ψ_B; the bulk is symmetric under the fold reflection); the heavy odd Ψ_A states, which carry a tip Z charge, decay through the oriented part of I9;
- its mass is an x_0 mass: $p^2/\varepsilon^2 = 0.896$, $m/m_H = 2.29$, and its hidden pressure $p_0/\rho = 0.772 = -d\ln\varepsilon_{DM}/d\ln f$ (two independent computations agree to 10^{-9}).

3.2 Relic abundance

Through the gauge-kinetic portal $S_{I12} = -\int\sqrt{|g|}\,\bar\Psi\Psi(c_G G^2 + c_W W^2 + c_B B^2)/\Lambda_\star^3$ the four-dimensional operator is $G_i\bar\chi\chi F^i_{\mu\nu}F^{i\mu\nu}$ with $G_i = \mathcal{O}\,c_i g_i^2/\Lambda_\star^3$, $\mathcal{O} = \partial\varepsilon_{DM}/\partial m = 0.7072$. The annihilation is p-wave: $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$ per abelian channel, $(M^4v^2/\pi)(8G_g^2+3G_W^2+G_B^2)$ in total. The thermal relic (Kolb-Turner freeze-out in a radiation-dominated Universe, Dirac fermion, $g_* = 56.75$) requires (DERIVED)

$$x_f = 30.31, \qquad \langle\sigma v\rangle(x_f) = 1.37\times10^{-25}\ \mathrm{cm}^3/\mathrm{s}, \qquad (8G_g^2+3G_W^2+G_B^2)^{1/2} = (16.6\ \mathrm{TeV})^{-3},$$

below the s-wave thermal unitarity estimate $(4\pi/M_{DM}^2)\sqrt{x_f/\pi} = 2.3\times10^{-25}$ cm^3/s (an s-wave estimate applied to a p-wave process). The portal scale is therefore of the order of the dark-matter mass, not far above it: the effective description is at its edge (a dedicated ultraviolet completion of I12 is OPEN). The relic value also assumes that the radion oscillation left over from the deflation, which redshifts like dust, does not dominate and dilute the Universe after freeze-out: it is conditional on the solution of the moduli problem.

3.3 Detection

- Direct (ESTIMATE): through gluons only, $\sigma_{SI} = \mu^2 f_N^2/\pi = 4.3\times10^{-53}$ cm^2 with $f_N = G_g\frac{8\pi}{9\alpha_s}m_N f_{TG}$, eight orders below the LZ limit extrapolated to 44 TeV (2.4×10^{-45} cm^2): the relic coupling could sit entirely in gluons (the limit would allow a gluon coupling whose square is 5.7×10^7 times the relic requirement). The electroweak channels have no tree-level nucleon coupling.
- Indirect: the annihilation is p-wave, $\propto v^2 \sim 10^{-6}$ in galaxies today, so present-day annihilation signals are suppressed by about 10^{-5} relative to freeze-out.
- Colliders: 44 TeV is beyond the reach of present and planned colliders.

3.4 The massless quanta are not dark matter

The fold bands of both halves are massless four-dimensional fermions ($\varepsilon = e^{-a_4}|k|$); a gas of them has $w = 1/3$ exactly (Kohn-Sham, DERIVED). The Ψ_A fold band couples to the Z through I9 and is part of the radiation ($N_{eff} = 3.044$, two of the three CNB species; it replaces ν_μ and ν_τ in the Z invisible width, $r^2 = 0.992$). The Ψ_B fold band must stay cold: if it were thermalised above the QCD crossover it would add $\Delta N_{eff} = 2(10.75/47.75)^{4/3} = 0.274$ (0.218 above the electroweak scale), which at fixed Planck parameters changes the CMB temperature spectrum by up to 4.2 % for $30 \le \ell \le 2500$ (a joint refit is OPEN).

3.5 Assessment

The association of dirac16complex with dark matter is valid in UM-2 under stated conditions: (i) the good sector (no extra-time momenta; outside it the halves mix and modes are tachyonic, OPEN); (ii) the exact fold parity of Ψ_B, which UM-2 derives from $P_+\Psi_B = 0$ at the tip; (iii) a gauge-kinetic portal I12 with coefficients set by the relic abundance; (iv) a Ψ_B fold band that never thermalises; (v) no late dilution by the radion (the moduli problem). It is the heavy odd part of one half of the field, not the field as a whole, that is dark matter; the field's massless fold bands are radiation.

4 dirac16complex as dark energy

4.1 No positive-energy state accelerates (DERIVED)

On the static block every computed Kohn-Sham state of Ψ has $w = \int p_3 / \int \rho \ge 0$: dark matter 0 ($2.8 \times 10^{-5}$ with λ_B), fold-band gas 1/3 (0.3312 with $+\lambda_B$, 0.3354 with $-\lambda_B$).

4.2 The condensate runs away (DERIVED)

A homogeneous condensate $S = \bar\Psi\Psi \propto a^{-3}$ with $U = \frac{\lambda}{2}S^2$ has $\rho = mS + U$, $p = SU' - U = \frac{\lambda}{2}S^2$, so

$$w(a) = \frac{x(a)}{1 + x(a)}, \qquad x(a) = \frac{\lambda S}{2m} = x_0 a^{-3}, \qquad x_0 = \frac{w_0}{1 - w_0}, \qquad w_a \equiv -\frac{dw}{da}\Big|_{a=1} = \frac{3x_0}{(1 + x_0)^2}.$$

$w < 0$ needs $\lambda < 0$, and then the energy density $\rho = mS(1 + x)$ crosses zero at $a = (-x_0)^{1/3}$:

w_0	-1.2	-1	-0.9	-0.861	-0.764	-0.5	-0.3
w_a (tangent)	-7.92	-6.00	-5.13	-4.81	-4.04	-2.25	-1.17
energy zero at z	0.224	0.260	0.283	0.293	0.322	0.442	0.630

The supernova fits of the author's private file, a constant $w = -0.764$ and $(w_0, w_a) = (-0.861, -0.60)$, are not reproduced: a condensate tuned to $w_0 = -0.861$ has $w_a = -4.81$ and negative energy density beyond $z = 0.29$.

4.3 The vacuum energy and the tuning

A natural vacuum energy of the cell, $m_{KK}^4/(16\pi^2) = 1.4 \times 10^{18}$ GeV^4, exceeds the dark-energy density (2.5×10^{-47} GeV^4) by 5.5×10^{64}. In UM-2 the four-dimensional cosmological term is the residual of the stabiliser balance, $\Lambda_4 = [6H^2k(1-2w_2k)+\Lambda+\kappa\rho_b]/G(k)$, zero at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$: the cosmological-constant problem, unsolved (OPEN). A slowly deflating block cannot provide a dynamical dark energy instead, because it would make α, G_N and the masses drift (the bound $|h| < 3.6 \times 10^{-8} H_0$ above).

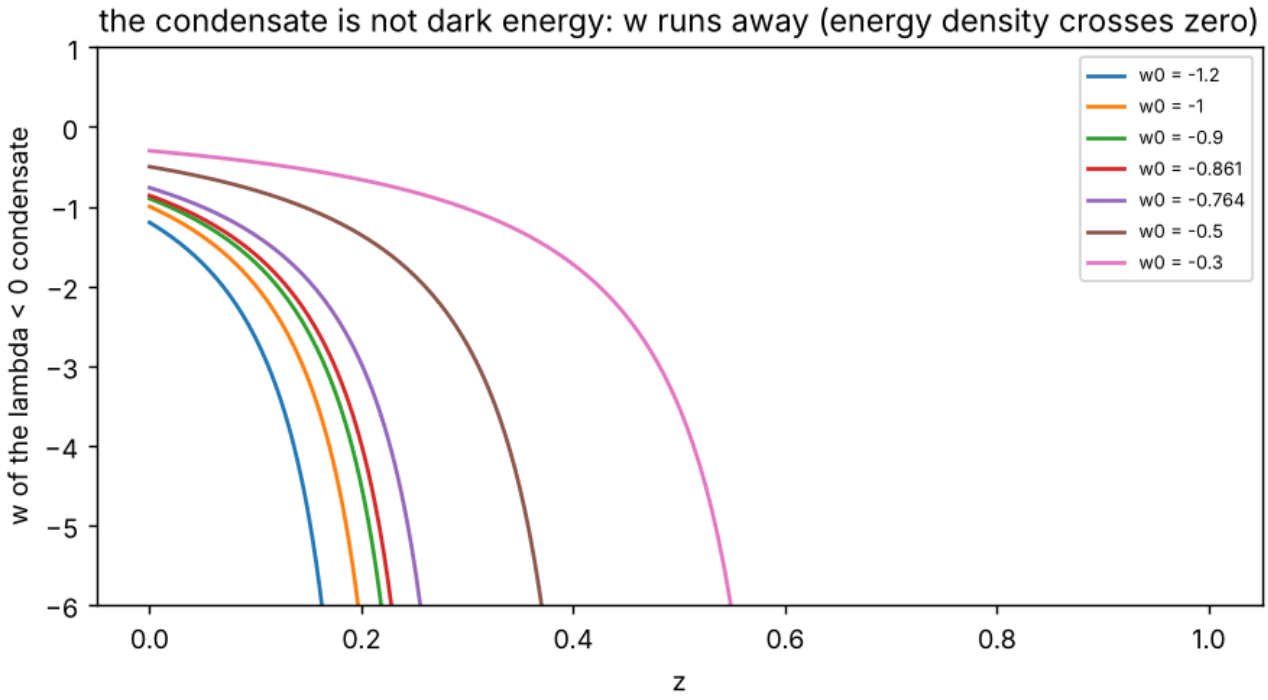


Figure 2: The equation of state of the $\lambda < 0$ condensate for several present values w_0: w runs away to $-\infty$ where the energy density crosses zero.

4.4 Assessment

The association of dirac16complex with dark energy is not supported in UM-2: its positive-energy states are matter or radiation, its only accelerating configuration is unstable and reaches negative energy density at $z \approx 0.2$ to 0.6, and the measured (w_0, w_a) are not reproduced. The dark energy of UM-2 is the tuned four-dimensional cosmological term Λ_4.

5 Conditions, open problems and conflicts

- **Conditions on which the conclusions rest** (ASSUMED): the stabiliser and its f^{-8} law; the good sector; the regular tip at $\eta = 3$; $m = H$; the tip sigma field and the boundary assignments of the Standard-Model fields; the portals I9, I11, I12 (INVENTED).
- **Open**: the origin of the stabiliser; the extra-time modes; the closed time-like curves implied by compact extra times; the radion moduli problem; the block-shape mode driven by $p_0 - p_t$; the ultraviolet completion of I12; the joint CMB and BAO fit of the variant with a thermalised Ψ_B fold band; the cosmological-constant problem.
- **Conflicts with data inherent in the order**: no Higgs boson (one is observed), perturbative unitarity of W^+W^- scattering lost near 1.2 TeV, and a one-generation Z width of 1.23 GeV against 2.4955 GeV.